

API Layer

The API layer of the OCR Worker acts as the entry point for incoming OCR jobs. This layer communicates exclusively through RabbitMQ.

- Component: RabbitConsumerService (Hosted Service)
- Subscribes to the RabbitMQ input queue (paperless.ocr.input)
- Deserializes incoming messages (MessageModel / DocumentOcrRequestedDto)
- Validates required fields (document ID, file name, storage path)
- Delegates processing to the business layer (OcrProcessingService)
- Optionally publishes result messages to the output queue (paperless.ocr.results) or triggers a REST callback to the main Paperless API for persistence

This layer encapsulates all messaging infrastructure and isolates business logic from RabbitMQ-specific code.

Business Layer (not yet implemented)

The business layer contains the domain logic for OCR.

- Component: OcrProcessingService
- Retrieves the document file from MinIO via IFileStoreClient
- Invokes the OCR engine (ITesseractClient / IOcrEngine) to extract text content
- Builds the OCR result model (OcrResult, ResultModel) with extracted text and metadata
- Publishes the result to RabbitMQ (paperless.ocr.results) or updates the main service through its REST API
- Handles domain errors such as “document not found” or “OCR failed”
- Performs logging of processing duration and status